

Association between dietary fiber intake and the folate status of a group of female adolescents.

The main objective of this study was to assess the association between dietary fiber intake and the folate status of Canadian female adolescents. We also assessed dietary folate intakes and evaluated the prevalence of biochemical folate deficiency in these subjects. Female adolescents aged 14-19 y (n = 224) were recruited and fasting blood samples were collected. Dietary intakes (3-d food record) were recorded and participants were classified as lactoovovegetarians, semivegetarians, or omnivores on the basis of food-consumption patterns assessed with food-frequency questionnaires. Fourteen percent, 17%, and 26% of lactoovovegetarians, semivegetarians, and omnivores, respectively, had dietary folate intakes below their predicted requirements; 1%, 4%, and 23%, respectively, had serum folate concentrations indicative of deficiency. Despite low dietary folate intakes and serum folate concentrations, few subjects had homocysteine concentrations indicative of deficiency, suggesting that the degree of folate depletion had not yet produced functional consequences. Most important, results suggest that the consumption of nonstarch polysaccharide is significantly associated with serum folate concentrations ($P < 0.001$). For each 1-g increase in nonstarch polysaccharide intake, a 1.8% increase in serum folate concentration is expected. In summary, we propose that an increase in nonstarch polysaccharide intake may promote the intestinal biosynthesis of folate, providing a complementary strategy to enhance the folate nutriture of humans.